

CORVALLIS 10 SSW, OR, USA

WMO: 726960

Lat: 44.4185N

Lon: 123.3257W

Elev: 312

StdP: 14.53

Time Zone: -8.00 (NAP)

Period: 06-19

WBAN: 04236

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.2	26.4	16.1	12.5	28.1	21.4	16.2	29.8	10.3	47.7	8.9	46.5	0.7	N/A	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.3	90.8	67.1	86.9	66.2	83.3	64.6	68.9	86.7	67.3	84.2	65.7	81.5	3.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	84.4	75.3	60.5	79.5	73.0	59.2	75.9	71.1	33.1	86.6	31.7	84.2	30.5	81.8	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	6.7	5.7	DB	17.3	98.6	5.7	3.5	13.2	101.1	9.9	103.2	6.7	105.2	2.6	107.7	(4)
			WB	17.4	71.6	6.1	0.9	13.0	72.3	9.5	72.8	6.0	73.3	1.6	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.2	40.1	42.4	45.8	49.2	55.2	60.4	66.7	67.0	61.8	52.8	45.1	39.3	(6)
(7)		DBStd	11.13	6.48	5.49	5.04	5.20	5.42	5.59	4.77	4.79	5.46	5.13	6.35	6.90	(7)
(8)		HDD50	1302	312	216	148	75	13	0	0	1	29	172	336	(8)	
(9)		HDD65	4906	773	632	594	474	308	159	35	32	125	379	599	796	(9)
(10)		CDD50	2108	4	5	19	51	173	314	516	527	355	115	24	5	(10)
(11)		CDD65	238	0	0	0	0	4	23	86	94	30	1	0	0	(11)
(12)		CDH74	4190	0	0	0	20	143	459	1525	1455	566	22	0	0	(12)
(13)		CDH80	1536	0	0	0	1	24	146	608	578	176	3	0	0	(13)
(14)	Wind	WSAvg	2.2	2.1	2.4	2.7	2.5	2.0	2.1	2.3	2.1	2.0	1.9	2.2	2.2	(14)
(15)	Precipitation	PrecAvg	46.0	7.4	5.1	5.2	3.3	2.5	1.4	0.3	0.4	1.3	4.2	6.7	8.3	(15)
(16)		PrecMax	82.1	17.0	15.2	10.0	8.4	5.7	3.2	1.2	1.9	7.0	11.0	16.7	18.0	(16)
(17)		PrecMin	22.9	1.5	0.6	1.1	0.9	0.0	0.1	0.0	0.0	0.0	1.0	1.3	1.7	(17)
(18)		PrecStd	11.7	4.1	3.3	2.3	1.8	1.5	0.8	0.3	0.6	1.4	2.1	3.6	3.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	59.7	67.8	77.3	82.9	91.1	96.3	96.1	89.9	76.7	64.0	58.1	(19)
(20)		MCWB	53.0	53.0	55.4	62.1	63.8	69.6	66.4	66.9	65.2	60.0	57.1	54.7	(20)	
(21)		2%	DB	54.2	56.3	62.1	69.8	78.3	84.1	91.0	90.7	85.3	71.2	60.0	55.0	(21)
(22)		MCWB	50.8	50.4	52.7	57.5	62.5	66.0	66.9	67.1	64.2	57.5	55.1	52.6	(22)	
(23)		5%	DB	51.9	53.4	58.8	64.2	73.4	79.0	86.7	86.5	80.6	66.5	57.2	52.0	(23)
(24)		MCWB	49.0	49.2	51.1	53.6	60.1	63.3	66.2	66.0	62.5	56.3	53.7	50.0	(24)	
(25)		10%	DB	49.8	51.1	55.5	59.8	68.6	74.8	82.5	82.3	76.1	62.7	54.8	49.2	(25)
(26)		MCWB	47.5	48.1	49.7	51.6	57.8	60.9	64.6	64.3	60.9	55.7	51.6	47.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	55.0	56.5	63.5	66.2	71.0	70.3	70.8	68.8	64.3	59.1	55.4	(27)
(28)		MCDWB	56.7	58.1	63.5	75.4	79.3	88.1	88.6	91.0	84.5	72.9	61.6	56.6	(28)	
(29)		2%	WB	52.3	52.6	54.5	58.4	63.4	67.6	68.4	68.6	66.5	61.3	56.5	52.6	(29)
(30)		MCDWB	54.1	55.4	60.1	68.1	74.9	83.2	87.0	87.1	80.8	67.4	58.8	54.5	(30)	
(31)		5%	WB	50.0	50.4	52.7	55.4	60.4	64.5	66.5	67.1	64.0	59.0	54.3	50.3	(31)
(32)		MCDWB	51.8	53.1	57.7	62.3	71.0	77.8	84.6	84.2	77.2	64.8	56.6	51.8	(32)	
(33)		10%	WB	48.2	48.5	50.7	52.8	58.4	62.1	64.9	65.4	62.1	57.1	51.9	47.5	(33)
(34)		MCDWB	50.2	51.0	55.3	58.4	67.7	73.9	81.6	81.3	73.6	61.5	54.3	49.1	(34)	
(35)	Mean Daily Temperature Range		MDBR	11.4	13.3	16.6	18.9	22.2	24.1	29.3	29.4	26.4	19.9	13.5	10.6	(35)
(36)		5% DB	MCDBR	13.2	16.4	22.6	27.6	31.9	32.5	36.5	36.5	35.2	27.6	15.4	12.0	(36)
(37)			MCWBR	9.8	10.8	14.1	16.0	17.9	16.1	15.6	15.1	15.7	15.8	11.1	10.0	(37)
(38)			MCDBR	12.1	13.0	20.2	24.7	29.0	30.3	34.6	33.9	31.1	21.2	13.3	11.2	(38)
(39)		5% WB	MCWBR	9.7	10.1	13.2	15.7	16.9	15.6	15.8	15.2	14.6	12.6	9.9	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.316	0.322	0.356	0.363	0.356	0.349	0.355	0.344	0.325	0.322	0.326	(40)	
(41)		taud	2.542	2.550	2.540	2.405	2.405	2.429	2.467	2.472	2.493	2.557	2.532	2.477	(41)	
(42)		Ebn at Noon	258	280	292	288	288	289	290	285	280	272	252	239	(42)	
(43)		Edh at Noon	21	25	29	35	36	36	34	33	30	24	21	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	420	674	989	1420	1805	2000	2277	1951	1469	869	474	339	(44)	
(45)		RadStd	61	85	118	116	170	176	135	121	121	97	61	49	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+31

Nomenclature: See separate page